Design Study of scFlowVis: Supporting Single-Cell RNA Sequencing Analysis through Visual Analytics

Supplementary Material III: Initial Online User Survey

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SCFlowVis User Survey

Key Information and Consent

You are invited to participate in this survey that will contribute to a research project taking place at King's College London (KCL) (Ethical Clearance Reference: MRSP-23/24-41723). Before you decide whether you want to take part, it is important for you to understand why the research is being done and what your participation will involve.

Please take the time to read the following information carefully and discuss it with others if you wish.

What is the purpose of the study?

The aim of the study is to develop and refine a visual analytics tool specifically designed for researchers and practitioners involved in single-cell RNA sequencing data analysis. We are focused on identifying the current challenges these professionals encounter, understanding their expectations from a visual analytics tool in aiding decision-making, facilitating data exploration and explanation, and revealing insights. Through this initiative, we seek to collect valuable feedback on the features of our current prototype to enhance its effectiveness in supporting the downstream analysis pipeline.

Why have I been invited to take part?

This is an anonymous survey. You can choose to participate voluntarily if you are over 18 years old and can use a computer. We invite you to take part in this research study because you have shown an interest in taking part in our study.

What will happen if I take part?

Upon agreeing to participate, you will initially answer questions regarding your background to help us understand your perspective better. Subsequently, you'll enter the requirement collection phase, where you'll be asked about your current workflow, specific domain needs, and expectations from the tool. After completing these sections, you'll evaluate the prototype's current features using a Likert scale questions. Finally, we'll inquire about your interest in participating in a follow-up user evaluation to further refine our tool.

Duration and Privacy:

The survey will take you approximately 10-15 minutes to complete. Your responses will be kept completely anonymous. Any data that we collect will be stored securely on the KCL servers. Demographic information gathering is for research analysis and will not be identifiable. We take your privacy very seriously.

Do I have to take part?

Participation is completely voluntary. You should only take part if you want to and choosing not to take part will not disadvantage you in any way. You are free to withdraw at any point during the completion of the survey, without having to give a reason by simply not submitting the study. We believe this study presents no risks to you. Once you submit the survey, it will no longer be possible to withdraw from the study because the data will be anonymous.

Contact Information:

If you have any questions, require more information about this study, or think you have been

harmed by participating please contact the primary researcher using the following contact details:
Yiwen Xing, Department of Informatics, King's College London, Bush House, 30 Aldwych, London WC2B 4BG, United Kingdom
Email: yiwen.xing@kcl.ac.uk

What if I have further questions, or if something goes wrong?
If this study has harmed, you in any way or if you wish to make a complaint about the conduct of the study you can contact KCL using the details below for further advice and information:
Dr. Alfie Abdul-Rahman, Department of Informatics, King's College London, Bush House, 30 Aldwych, London WC2B 4BG, United Kingdom
Email: alfie.abdulrahman@kcl.ac.uk

Please note that this survey will be best displayed on a laptop or desktop computer. Some features may be less compatible for use on a mobile device.

Thank you!

By checking the box, you are confirming that you are 18 years or older, have read and understood the above information, and agree to take part in the research. Your participation is voluntary, and you are free to leave at any time by simply closing the web browser.

○ I agree

Part 1. Participant Background

Do you work in a field related to bioinformatics?

○ Yes
○ No

How long have you been involved in the bioinformatics domain?

Less than a year	1-3 years	3-5 years	5-7 years	More than 7 years
○	○	○	○	○

To what extent is your work related to single-cell RNA sequencing (scRNA-seq) analysis?

Very low	Low	Medium	High	Very High
○	○	○	○	○

Which platforms do you frequently utilize for data analysis in your work? (Please select all that apply)

☐ Web/Cloud-based platforms
☐ Desktop applications

- [ ] Python (including notebooks)
- [ ] R (including markdown and scripts)
- [ ] _____ Others (please specify)

Are you familiar with any of the following APIs, systems, or software? Please check all that apply and add any others not listed.

- [ ] Scanpy
- [ ] Seurat
- [ ] ASAP
- [ ] cellxgene
- [ ] scViewer
- [ ] Loupe Browser
- [ ] SingleCellExperiment
- [ ] Monocle
- [ ] Others (please specify) _____

Have you used any of the following APIs, systems, or software for your work? If yes, please check all that apply. If you've used others not mentioned here, please kindly list their names.

- [ ] Scanpy
- [ ] Seurat
- [ ] ASAP
- [ ] cellxgene
- [ ] scViewer
- [ ] Loupe Browser
- [ ] SingleCellExperiment
- [ ] Monocle
- [ ] Others (please specify) _____

How often do you incorporate visualization into your workflow?

Very rarely	Rarely	Occasionally	Frequently	Very frequently
○	○	○	○	○

What are the primary purposes of using visualization in your workflow? (Please select all that apply)

- [ ] Supporting decision-making
- [ ] Exploring data
- [ ] Validating hypotheses
- [ ] Revealing insights
- [ ] Other purpose (please specify)
- [ ] Explaining data
- [ ] Presenting data to others
- [ ] Recording actions

Part2. Domain User's Expectation of the Tool

Do you think it would be beneficial for a tool to support the following functions throughout the entire analysis pipeline?

	Yes	No
Managing AnnData Dynamics during Analysis	○	○
Recommending Suitable Visualization Methods	○	○
Documenting All Operations and Parameter Settings	○	○

	Yes	No
Comparing Different Results from Different Operations	○	○
Presenting and Sharing the Analysis Flow	○	○
Reducing the Coding Workload for Conducting Data Analysis	○	○
Reducing the Coding Workload for Generating Proper Visualizations	○	○

Do you think the **existing tools** can support the following functions throughout the entire analysis pipeline?

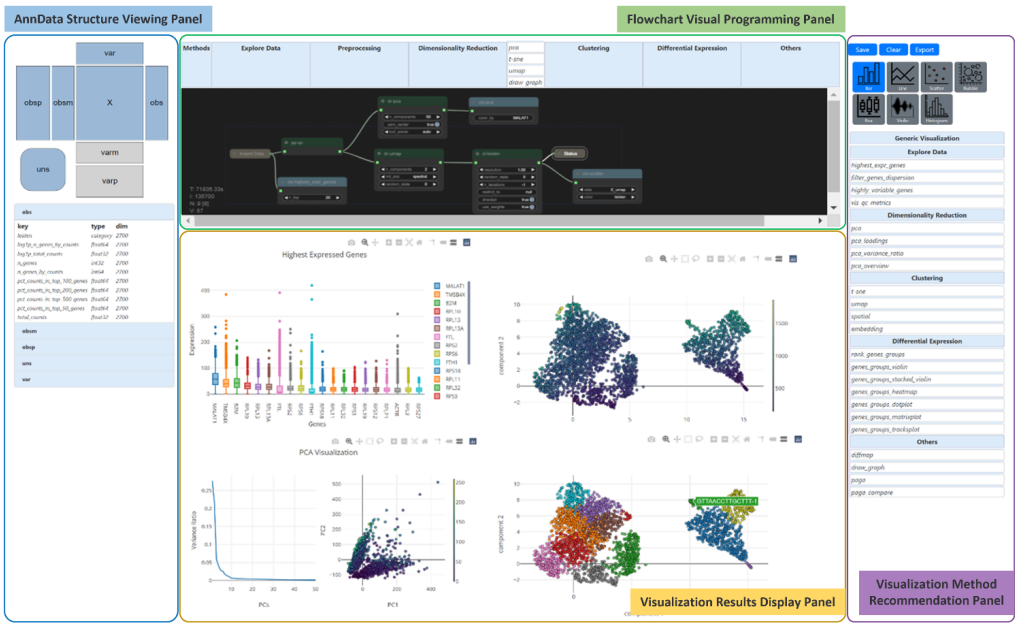
	Yes	No
Managing AnnData Dynamics during Analysis	○	○
Recommending Suitable Visualization Methods	○	○
Documenting All Operations and Parameter Settings	○	○
Comparing Different Results from Different Operations	○	○
Presenting and Sharing the Analysis Flow	○	○
Reducing the Coding Workload for Conducting Data Analysis	○	○
Reducing the Coding Workload for Generating Proper Visualizations	○	○

Do you think the **existing tools** can meet the requirements commonly encountered in the single-cell RNA sequencing (scRNA-seq) downstream analysis pipeline? If you have other requirement suggestions, please specify.

	Yes	No
Understanding High-Dimensional Data	○	○
Handling Batch Effects and Data Integration	○	○
Identifying and Interpreting Heterogeneity	○	○
Estimating and Validating Cell Type Markers	○	○
Gene Function Annotation and Enrichment Analysis	○	○
Others (please specify)	○	○

If you feel that the **existing tools** do not fully satisfy the above requirements, or there is room for improvement, please describe the functionalities you expect to see enhanced or introduced. For example, improvements in tool-user interaction, etc.

Part 3. Feedback on the Prototype's Functionalities



In this section, we seek your valuable feedback on the helpfulness of four specific functionalities to be implemented in our visual analytic tool. Each functional panel will be briefly described followed by a question asking you to rate its helpfulness on a 5-point Likert scale.
Please note, the interface displayed below includes visual demonstrations of the four functional panels in question.

○ Understand, ready to go.

Flowchart Visual Programming Panel allows users to manipulate data through the addition of nodes for processes such as preprocessing, dimension reduction, clustering, correlation analysis, and visualization. Each step is recorded as nodes and links on the panel. After completing the analysis, users can save the flowchart or workspace and share it with colleagues.

How helpful do you find this panel for your work?

Unhelpful	Somewhat unhelpful	Neutral	Somewhat helpful	Very helpful
○	○	○	○	○

AnnData Structure Viewing Panel offers a thumbnail and list view of the AnnData structure, allowing users to dynamically grasp the changes in AnnData content after each data operation and view the types and dimensions of all data stored in various AnnData objects.

How helpful do you find this panel for understanding data structure and the effects of your operations?

Unhelpful	Somewhat unhelpful	Neutral	Somewhat helpful	Very helpful
○	○	○	○	○

Visualization Method Recommendation Panel allows users to search for suitable visualization methods based on their known objectives or choose from system-recommended visualization methods based on the current data attributes.

How helpful do you find this panel in finding appropriate methods for visualizing your data?

Unhelpful	Somewhat unhelpful	Neutral	Somewhat helpful	Very helpful
○	○	○	○	○

Visualization Results Display Panel displays the visualization results generated during the analysis processus. Users can interact with the data through mouse clicking, dragging, and hovering to further filter, zoom in/out, and other operations. Moreover, multiple visualizations can be arranged freely on this panel for result comparison purposes.

How helpful do you find this panel for showing analysis results and interacting with the data?

Unhelpful	Somewhat unhelpful	Neutral	Somewhat helpful	Very helpful
○	○	○	○	○

Part 4. Willingness to the Follow-up Interview

Thank you for answering the above questions. We will conduct a user experience study for this tool soon, and we would appreciate it if you could take part.

Would you be happy for us to contact you in future about participating in this upcoming study?

○ Yes
○ No

Thank you for your willingness to participate in the next survey.
Please leave your email address below and we will get in touch with you again.

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